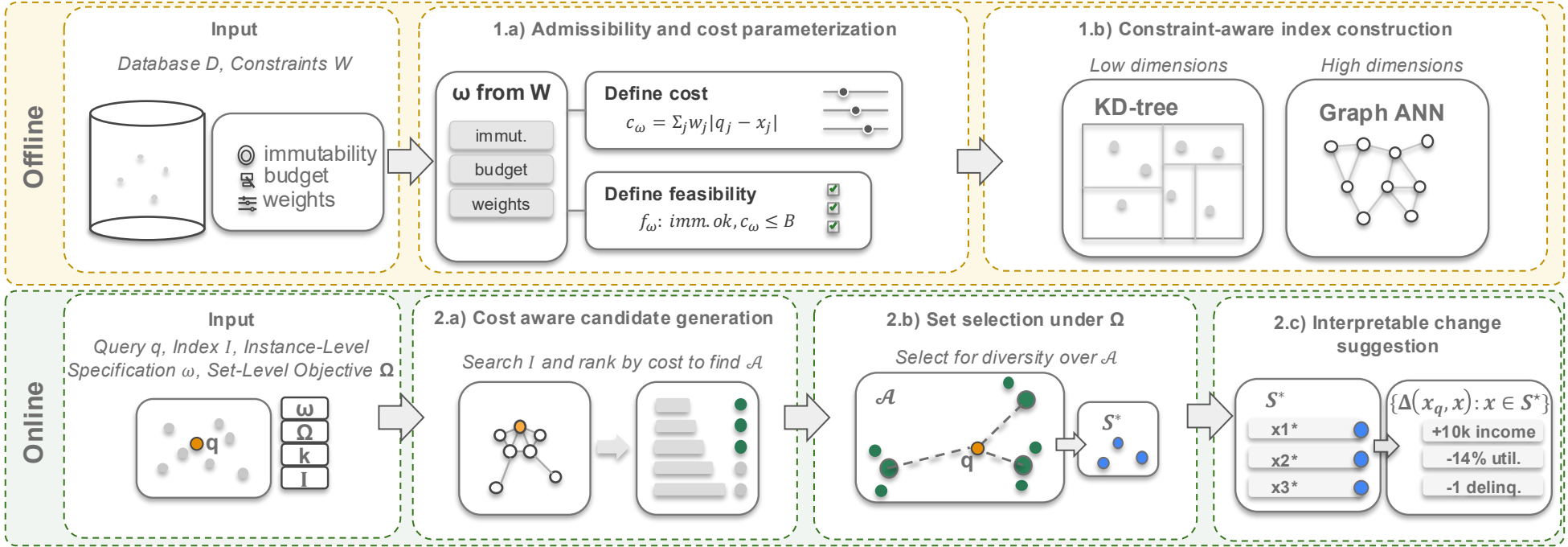

Offline
Input
Database D, Constraints W
immutability
budget
weights
1.a) Admissibility and cost parameterization
ω from W
immut.
budget
weights
Define cost
$c_\omega = \Sigma_j w_j |q_j - x_j|$
Define feasibility
f_ω: $imm.ok, c_\omega \leq B$
1.b) Constraint-aware index construction
Low dimensions
High dimensions
KD-tree
Graph ANN
Online
Input
Query q, Index I, Instance-Level Specification ω, Set-Level Objective Ω
q
ω
Ω
k
l
2.a) Cost aware candidate generation
Search I and rank by cost to find $\mathcal{A}$
2.b) Set selection under Ω
Select for diversity over $\mathcal{A}$
$\mathcal{A}$
q
S^*
2.c) Interpretable change suggestion
S^*
x1*
x2*
x3*
$\{\Delta(x_q, x) : x \in S^*\}$
+10k income
-14% util.
-1 deling.